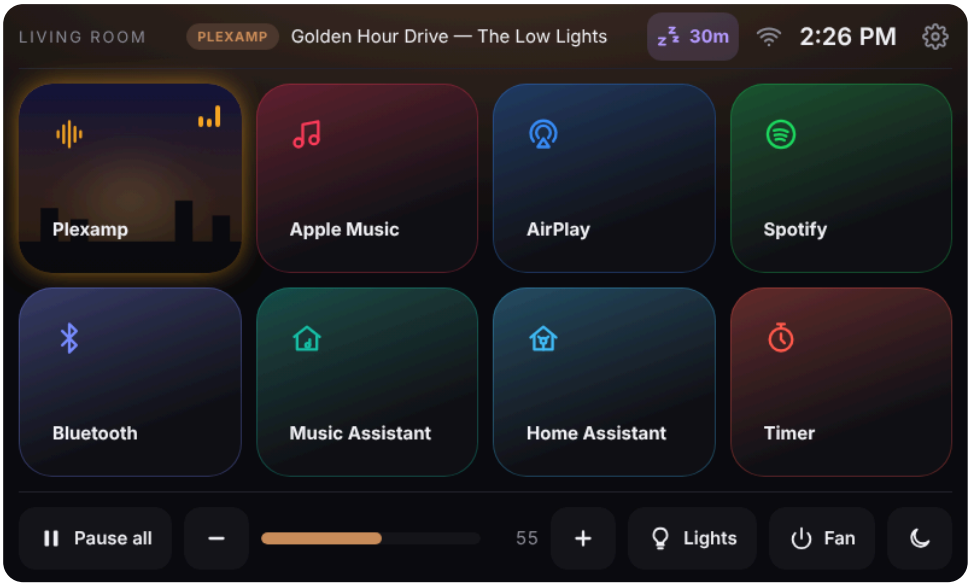
PlexPie

Setup guide



github.com/ROGUEONE-SZ/PlexPie

Build and install

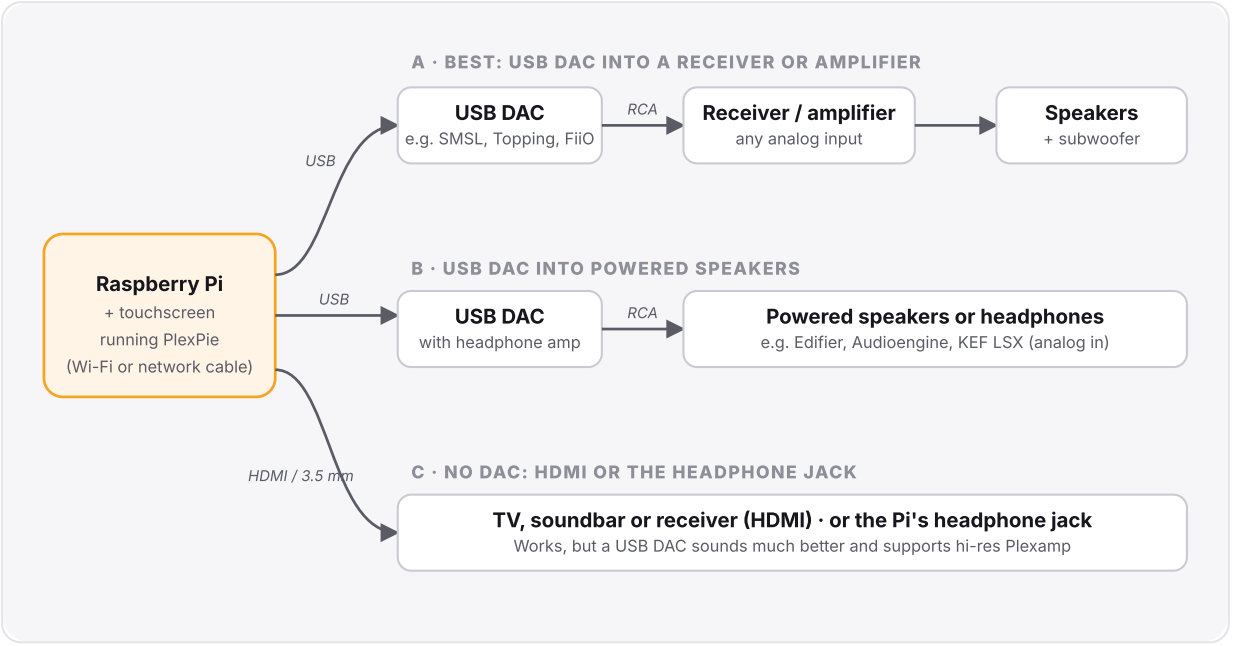
From a box of parts to music playing, in about an hour. You only need a computer for the first part.

1. What you need

Part	Notes
Raspberry Pi 4 or 5	2 GB or more. A Pi 4 with 4 GB is plenty.
Official power supply	Cheap phone chargers cause random freezes.
microSD card	32 GB or larger, "A1" or "A2" rated (SanDisk, Samsung).
Touchscreen	The official 7" Touch Display or Touch Display 2, or any HDMI touchscreen.
USB DAC (recommended)	From about $50 (SMSL, Topping, FiiO). Sounds far better than the headphone jack.
Cable	RCA (DAC to receiver) or 3.5 mm (DAC to powered speakers).
Accounts	Plex (free). Optional: Apple Music, Spotify Premium, Home Assistant.

2. Connect the hardware (power unplugged)

1. **Screen:** the official display's ribbon cable goes in the Pi's DISPLAY connector. An HDMI screen uses HDMI plus its USB cable (for touch).
2. **Network:** a network cable is most reliable. Wi-Fi works too.
3. **Sound:** pick one of these:



3. Put Raspberry Pi OS on the memory card

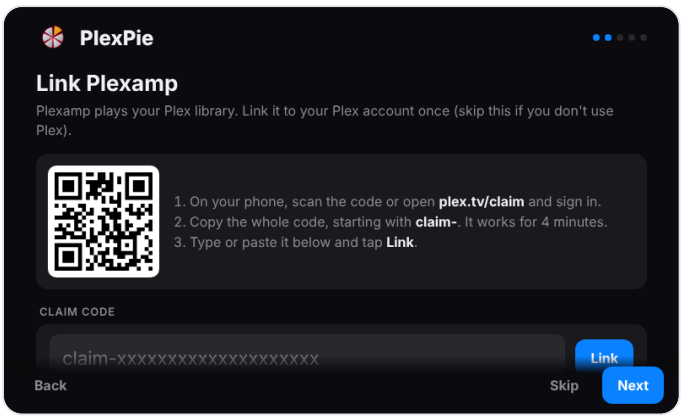
1. On your computer, install **Raspberry Pi Imager** from `raspberrypi.com/software` and open it.
2. `Choose Device` : your Pi. `Choose OS` : **Raspberry Pi OS (64-bit)**. `Choose Storage` : the memory card.
3. Click `Next` , then `Edit Settings` : set a **hostname** (e.g. `plexpie`), a **username and password** (write them down), your **Wi-Fi** if you're not using a cable, and your time zone.
4. On the `Services` tab, turn on **Enable SSH** with password authentication.
5. `Save` › `Yes` › `Yes` . When it's done, put the card in the Pi and plug in the power. The first start takes 2 to 3 minutes.

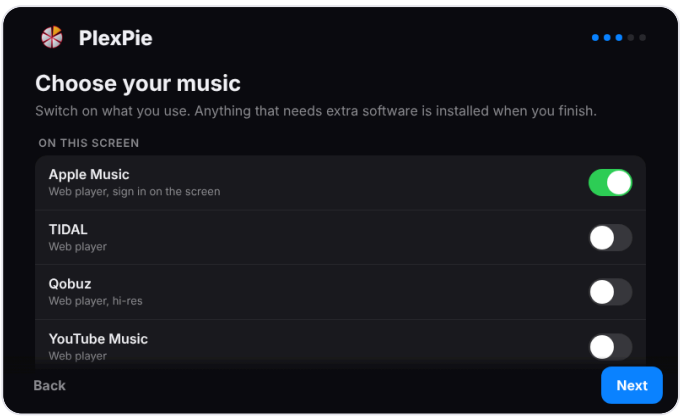
4. Install PlexPie

1. On your computer, open **Terminal** (Mac: ⌘ Space, type Terminal) or **PowerShell** (Windows: Start, type PowerShell).
2. Type `ssh pi@plexpie.local` (your username and hostname), press Enter, type `yes` , then your password. Nothing shows while you type the password; that's normal.
3. Paste this line and press Enter:

```
curl -fsSL https://raw.githubusercontent.com/ROGUEONE-SZ/PlexPie/main/get.sh | ba
sh
```

4. Press `Enter` for **Quick setup**: it installs everything, then a **setup wizard** on the screen asks for the name, links Plexamp (scan the QR code on your phone), and lets you pick your apps. Or choose **2** to answer the questions in the terminal.
5. With Quick setup you're done here: follow the wizard on the screen (or on your phone at the address it shows). The next step is only for option 2.
6. When it asks for a **claim code**: open `plex.tv/claim` on any device, copy the code, paste it, press Enter, type the player name, and press Ctrl+C when Plexamp says it's ready.
7. It takes 15 to 30 minutes. When it's done, the screen shows the PlexPie home screen.
8. Screen standing up instead of lying down (the Touch Display 2 starts in portrait)? Settings › Screen › **Rotation** › Right or Left.



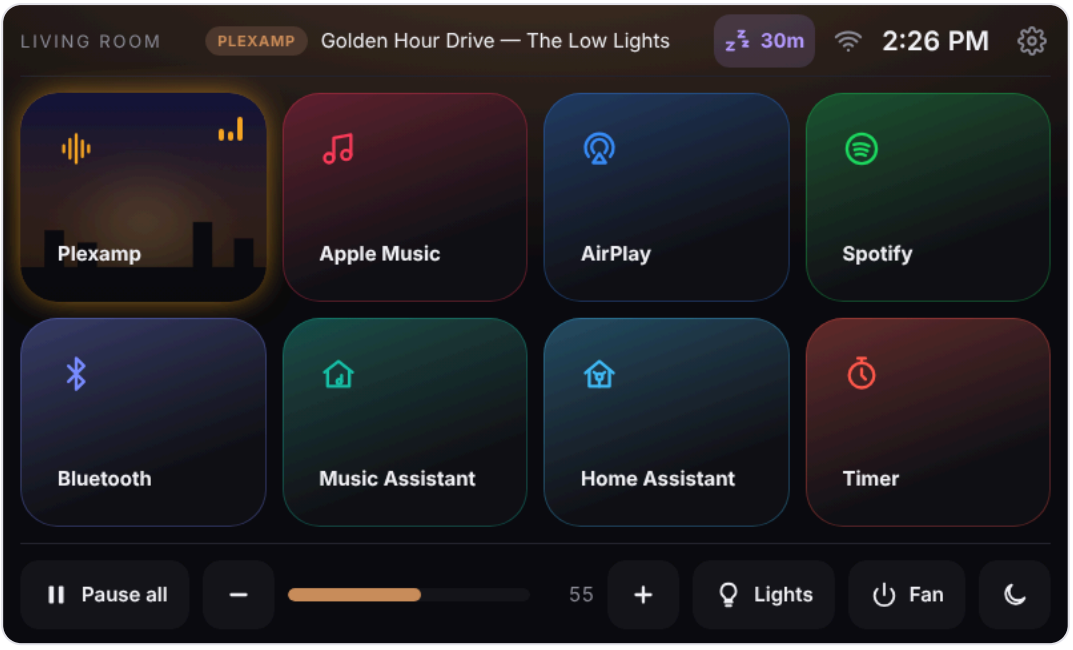


Can't connect? Use the Pi's number address instead of `plexpie.local`. Your router's list of devices shows it, e.g. `ssh pi@192.168.1.42`.

Plexamp

Your own music library, in full quality.

1. Plexamp is linked to your Plex account during the install (the claim code step). Settings › Configuration › Plexamp shows which account.
2. On the screen, tap **Plexamp** and pick music, or
3. On your phone or tablet, open the **Plexamp** app, tap the speaker icon and pick **your player**. Your phone becomes the remote.



Music plays but the screen spins or shows no "now playing"? Settings › Configuration › Plexamp › **Reset the Plexamp screen**, then sign in to Plex once on the screen. This happens after the player's network address changes.

Linked to the wrong account, or not linked? Run the installer again from a computer (Build and install, step 4). It asks for a new claim code.

Jellyfin

If you run a Jellyfin server, its music library opens right on the screen.

1. Settings › Configuration › **Jellyfin**: switch it on, enter your server's **address** (e.g. `192.168.1.20`, or with its port `192.168.1.20:8096`) and tap `Save`.
2. On the home screen, tap `Jellyfin` and sign in with your Jellyfin user.
3. Browse and play. Pause all, the track name and the cover work like the other apps.

> **Play from the Jellyfin app on your phone:** switch on **UPnP / DLNA** as well. With Jellyfin's DLNA plugin installed on your server, the player shows up as a device to play to. AirPlay and Bluetooth work too.

Apple Music

The Apple Music web player, right on the screen.

1. Settings › Configuration › **Apple Music**: switch it on and tap **Save**.
2. On the home screen, tap **Apple Music**, then **Sign In** (top right).
3. Enter your Apple ID. If you get a code on your iPhone, type it on the screen.
4. You stay signed in. PlexPie never sees or stores your password.

> **Want to play from your iPhone instead?** Use AirPlay: it works with Apple Music and every other app.

To sign out: Settings › Configuration › Apple Music › **Sign out**.

YouTube Music

YouTube Music's web player on the screen, like Apple Music.

1. Settings › Configuration › **YouTube Music**: switch it on and tap **Save**.
2. On the home screen, tap **YouTube Music**. It plays without an account (with ads); tap **Sign in** for your library, or Premium for no ads.
3. You stay signed in. PlexPie never sees or stores your password.

> You can't send music to it from the YouTube Music app on your phone (YouTube doesn't allow that for players like this). Use AirPlay or Bluetooth from your phone instead.

BBC Sounds

BBC radio stations, shows and podcasts on the screen.

1. Settings › Configuration › **BBC Sounds**: switch it on and tap **Save**.
2. On the home screen, tap **BBC Sounds** and sign in with your BBC account.
3. Pick a station or podcast. Pause all and the volume work as usual.

> Live radio plays in most countries; many shows and podcasts are only available in the UK.

TIDAL

TIDAL's web player on the screen.

1. Settings › Configuration › **TIDAL**: switch it on and tap **Save**.
2. On the home screen, tap **TIDAL** and sign in with your TIDAL account.

> TIDAL Connect (picking this player in the TIDAL app on your phone) is only licensed to hardware makers, so it isn't available here. To play from your phone, use AirPlay or Bluetooth.

Qobuz

Qobuz's web player on the screen, including hi-res albums.

1. Settings › Configuration › **Qobuz**: switch it on and tap **Save**.
2. On the home screen, tap **Qobuz** and sign in with your Qobuz account.

> Qobuz Connect (playing from the Qobuz app on your phone) isn't open to projects like this yet. To play from your phone, use AirPlay or Bluetooth.

Your own web players

Any music, radio or audiobook website can get its own tile, like Apple Music.

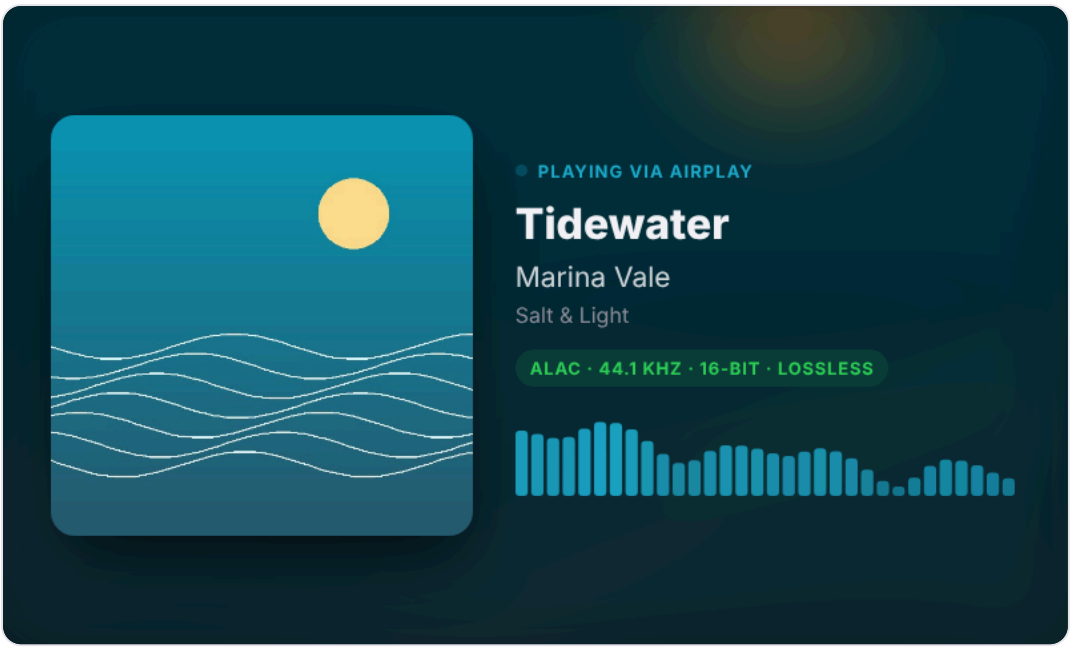
1. Settings › Configuration › **Your web players**.
2. Type a **Name** (shown on the tile) and the **Web address**, e.g. `https://tunein.com`. Tap **Add**, then **Save**.
3. The new tile appears on the home screen. Tap it and sign in on the screen if the site needs it.

> Pause all, the track name and the cover work on most sites. A few block sign-in from kiosk browsers or ask you to use their app; those can't be helped from PlexPie. Up to six web players.

AirPlay

Play anything from an iPhone, iPad or Mac.

1. Make sure your device is on the same Wi-Fi as the player.
2. **iPhone or iPad:** open Control Center (swipe down from the top-right corner), tap the AirPlay icon on the music tile, and pick **your player**.
3. **Mac:** click the AirPlay or Control Center icon in the menu bar and pick **your player**.
4. The screen shows the cover, title and a visualizer. Volume follows your device.



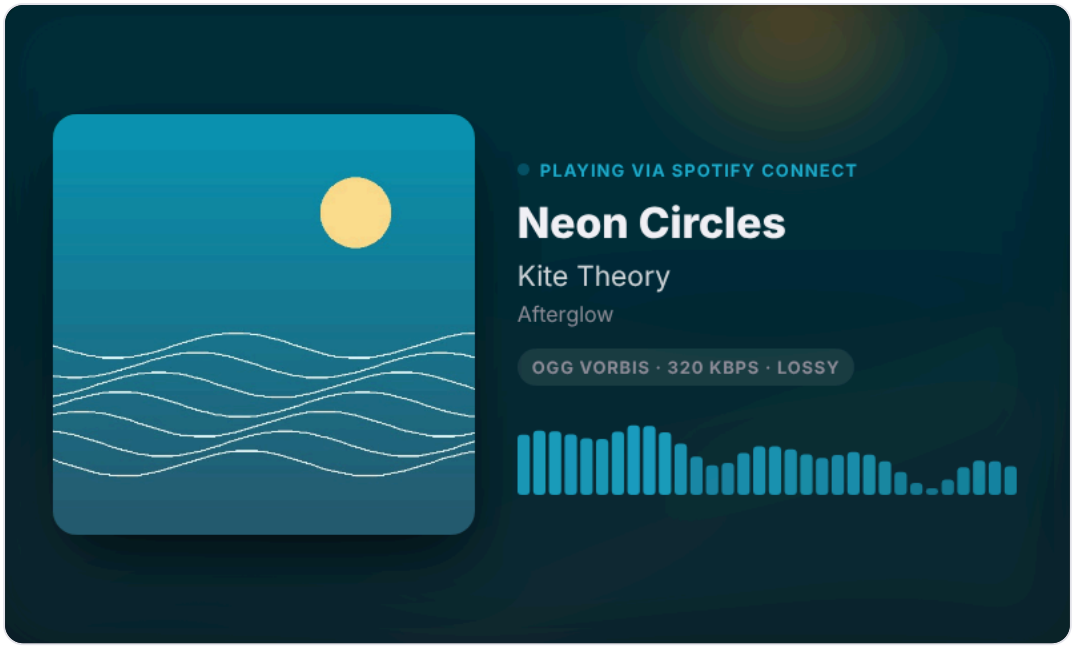
Don't see it? Check Settings › Configuration › AirPlay is on, and that your device isn't on a guest Wi-Fi.

Spotify

Two ways: pick the player from the Spotify app (Premium), or browse Spotify on the screen.

Spotify Connect (from your phone, tablet or computer)

1. Settings › Configuration › Spotify: switch on **Spotify Connect** and tap `Save`.
2. In the Spotify app, start a song, tap the **devices** icon (bottom left) and pick **your player**.



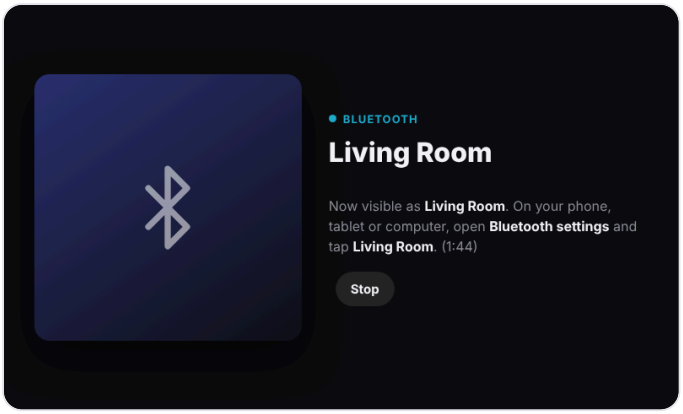
Spotify web player (on the screen)

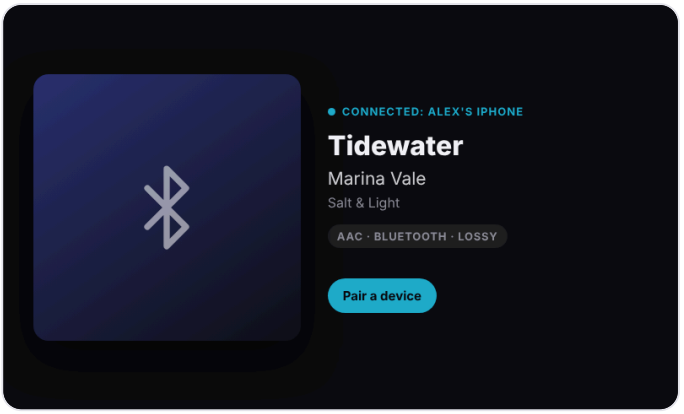
1. Switch on **Spotify web player** and tap `Save`.
2. Tap the Spotify tile on the home screen and log in once.

Bluetooth

Any phone, tablet or computer can play to it, even a guest's Android phone.

1. On the screen, tap **Bluetooth** › **Pair a device** (or Settings › Configuration › Bluetooth › **Pair**). The player is visible for 2 minutes.
2. On your device, open **Settings › Bluetooth** (Mac: System Settings › Bluetooth) and tap **your player**.
3. Confirm on your device if it asks. No PIN is needed.
4. Play music. Next time it connects by itself: just pick it in your device's Bluetooth or AirPlay/output list.





To remove a device: Settings › Configuration › Bluetooth › **Forget**.

UPnP / DLNA

The player becomes a standard network speaker (a "renderer") that many apps can play to.

1. On the Pi, run `bash configure.sh`, answer **yes** to **UPnP / DLNA**, then `sudo bash instal l.sh`.
2. Settings › Configuration › **UPnP / DLNA** shows it running.
3. In an app like **BubbleUPnP** (Android), **mconnect** (iPhone), Jellyfin, or your NAS's music app (Synology, QNAP…), pick **your player** as the speaker or renderer.

> BubbleUPnP and mconnect can also stream TIDAL and Qobuz to it, a way to play those from your phone. The screen shows the track and cover when the app sends them.

Music Assistant

If you run Music Assistant (usually inside Home Assistant), the player becomes one of its speakers.

1. Settings › Configuration › Music Assistant: switch it on.
2. **Server address:** the computer that runs Music Assistant, e.g. `homeassistant.local` or `192.168.1.20`.
3. **Web address:** the Music Assistant page, usually the same address with `:8095`, e.g. `http://homeassistant.local:8095`.
4. Tap `Save`. In Music Assistant, open Settings › Players: **your player** appears there. Turn it on.

> The dot next to **Server** turns green when the player can reach Music Assistant.

Caldera Music

Caldera Music is a free Plex Labs player. With it, the player shows up in Plexamp's speaker list and can play in sync with other Caldera speakers around the house.

1. On the Pi, run `bash configure.sh` and answer **yes** to **Add Caldera Music**, then `sudo bash install.sh`.
2. The installer downloads Caldera Music and asks you to sign in to Plex once. Follow what it shows on the screen.
3. In **Plexamp** on your phone, tablet or computer, tap the speaker icon and pick **your player (Caldera)**. To play in more rooms, add your other Caldera speakers there too.
4. When it plays, the screen shows **Playing via Caldera**. `Pause all` and the volume work as usual.

> You'll see the player twice in Plexamp: **your player** is Plexamp on the screen, **(Caldera)** is the multi-room one. Switch Caldera off any time in Settings › Configuration › Caldera Music. It shares the sound output with the other apps; for Caldera alone on the DAC, set `CALDERA_DEVICE="caldera"` in plexpie.conf and run the installer again.

Roon

With Roon Bridge, the player becomes a speaker in Roon. You need a Roon subscription and a Roon Server (Core) on your network.

1. On the Pi, run `bash configure.sh`, answer **yes** to **Roon**, then `sudo bash install.sh`. It downloads Roon Bridge from Roon Labs.
2. In the Roon app, open **Settings › Audio**. Under the player's name, **Enable** the output: **System Default** shares the DAC with the other apps; the DAC's own name is bit-perfect but takes the DAC while Roon plays.
3. Pick the new zone in Roon and play. The screen shows **Playing via Roon**; track details stay in the Roon app.

> **Pause all** stops Roon by restarting the bridge for a moment.

Snapcast

If you run a Snapcast server (on its own, or with Music Assistant or Mopidy), the player plays in sync with your other Snapcast speakers.

1. On the Pi, run `bash configure.sh`, answer **yes** to **Snapcast**, then `sudo bash install.sh`.
2. It finds the server on your network by itself. If not, enter its address in Settings › Configuration › **Snapcast**.
3. Put the player in a group in your Snapcast app (Snapweb, Snapdroid). The screen shows what the group plays.

> **Pause all** mutes this player in Snapcast; tap the Snapcast tile to unmute it.

Home Assistant

Control the player from Home Assistant, show a dashboard on the screen, and add your own buttons.

1. Connect (MQTT)

1. In Home Assistant, install the **Mosquitto broker** add-on (Settings › Add-ons) and the **MQTT** integration, if you haven't.
2. Create a Home Assistant user for the player (Settings › People › Users), e.g. `plexpie`.
3. On the player: Settings › Configuration › Home Assistant. Enter the **Broker** (your Home Assistant address), port `1883`, and that username and password. Tap **Save**.
4. In Home Assistant, the player appears under Settings › Devices & services › MQTT with now playing, volume, source, sleep timer, screen, brightness, pause/play, the PlexPie and Raspberry Pi OS updates, restart and shut down.

2. Dashboard tile on the screen

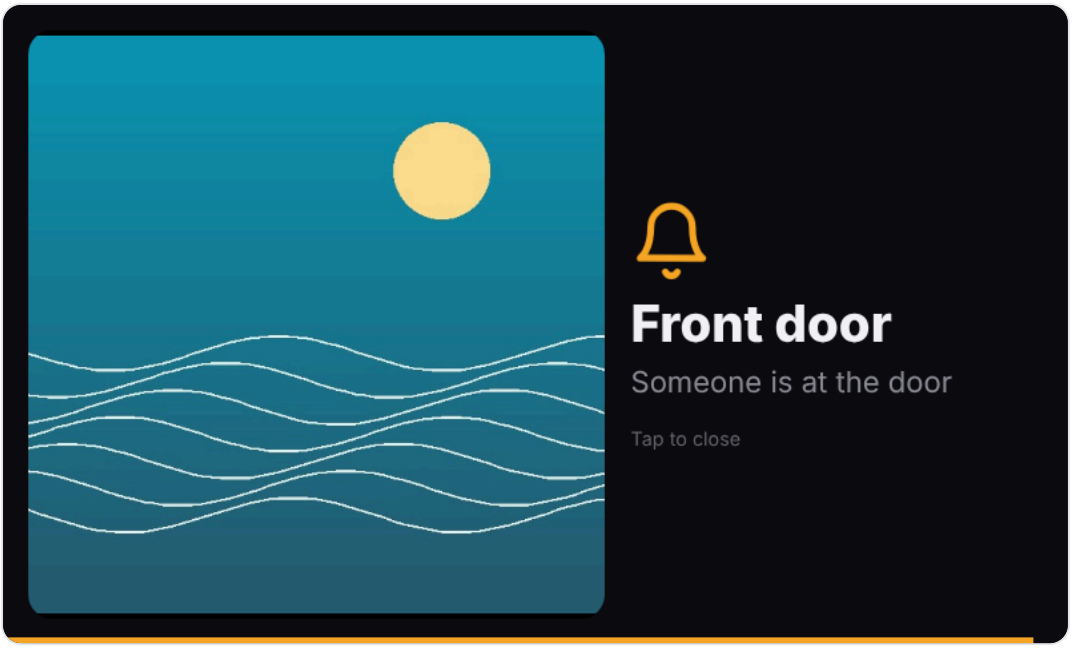
1. Switch on **Home Assistant tile** and enter the **Address**, e.g. `http://homeassistant.local:8123`.
2. Optional: make a small dashboard just for this screen and enter its name under **Dashboard**, e.g. `gym-screen`.
3. Tap the tile once and sign in to Home Assistant on the screen.

3. Launcher buttons

1. Under **Launcher buttons**, type names separated by commas, e.g. `Gym lights, Sound`, and tap **Save**.
2. Each one appears on the home screen and as a switch in Home Assistant. Use it in an automation, e.g. when "Gym lights" turns on, turn on the lights.

4. Alerts on the screen

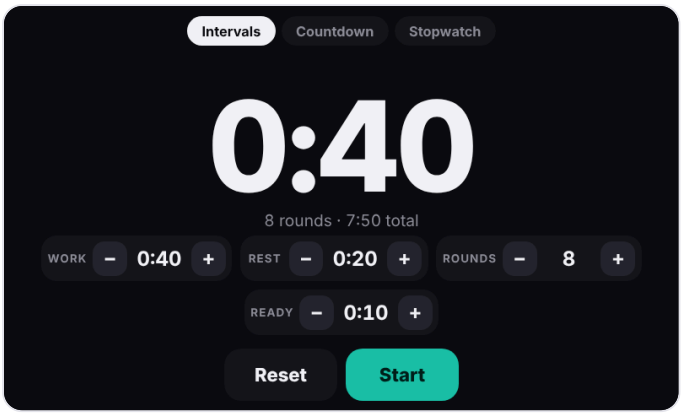
Call `notify.send_message` on the player's **Screen alert** entity with a message, e.g. "Someone's at the door". Add a camera picture with the MQTT topic `plexpie/<player>/alert/set`.

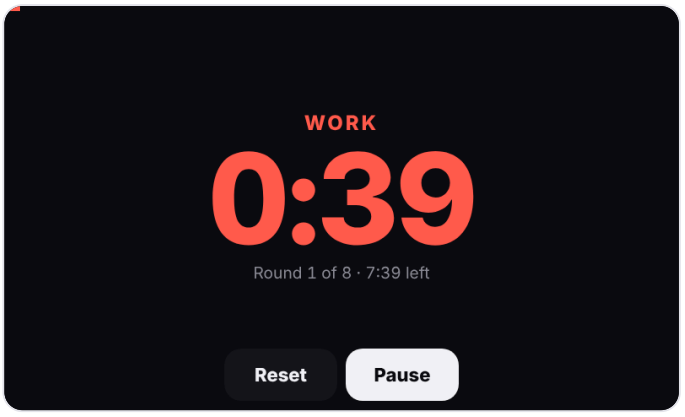

Front door
Someone is at the door
Tap to close

Workout timer

Big, readable, with beeps. Keeps the screen on while it runs.

1. Settings › Configuration › Workout timer: switch it on.
2. On the home screen, tap **Timer**. Pick **Intervals**, **Countdown** or **Stopwatch**.
3. For intervals, set Work, Rest, Rounds and Ready with − and + (hold to go faster), then **Start**.
4. It beeps for the last 3 seconds of each part. Music keeps playing.





Equalizer

Shape the sound for your room, speakers or headphones.

1. Settings › Sound: switch on **Equalizer**.
2. Pick a **Preset**: Less bass, More bass, Loudness (for quiet listening), Vocal, Soft treble, or Custom (5 sliders).
3. The curve shows what changes. Tap **Save**.

Headphones or room correction

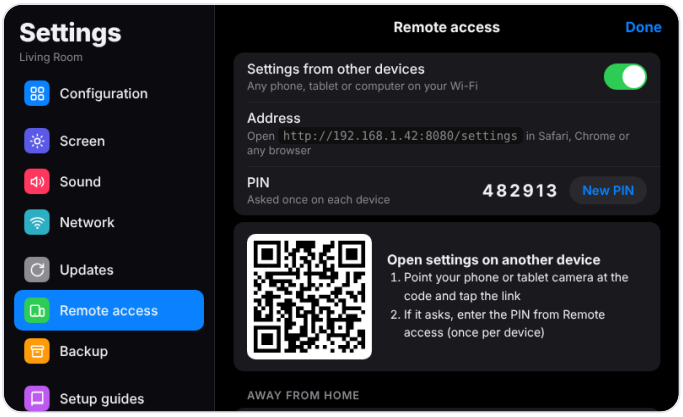
1. For headphones, open `autoeq.app`, pick your model, choose **Equalizer APO** and copy the parametric EQ.
2. For a room, measure with REW and export the filters as Equalizer APO text.
3. Pick the preset **Pasted filters** and paste the text.

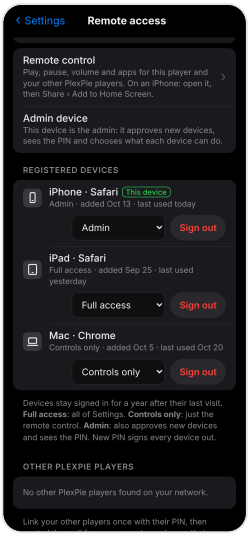
> With the equalizer off, lossless music plays untouched.

Remote access

Change settings from any phone, tablet or computer on your Wi-Fi.

1. On the player: Settings › **Remote access**. Note the **Address** and **PIN**, or scan the code with your camera.
2. On your device, open `the address` in Safari, Chrome or any browser.
3. Enter the PIN. You only do this once per device.
4. Tip: add it to your home screen (Safari: Share › Add to Home Screen) so it opens like an app.





Every device you sign in shows under **Registered devices**, with when it was last used. Tap **Sign out** to remove one, or **New PIN** on the player to sign them all out.

Your admin phone

1. Sign in on your own phone with the PIN, then Settings › Remote access › **Make this my admin device**.
2. From now on, a new phone or tablet can tap **Ask the admin** instead of typing the PIN. You get **Allow**, **Controls only** or **Deny** on your phone and on the player's screen.
3. Each device has a level: **Admin** (everything, sees the PIN), **Full access** (all of Settings) or **Controls only** (just the remote: play, pause, volume, apps). Change it next to the device under Registered devices.

One remote for all your players

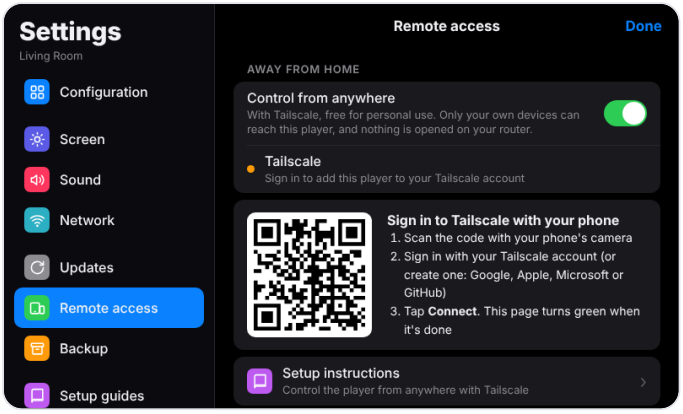
1. On your admin phone, Settings › Remote access › **Other PlexPie players** lists the players on your network. Type a player's PIN and tap **Link**, once per player.
2. Open **Remote control** (`/remote`) and add it to your home screen. Every linked player shows with what's playing, play/pause, volume and its apps.
3. **Settings ›** on any player opens its Settings on your phone straight away, without its PIN.

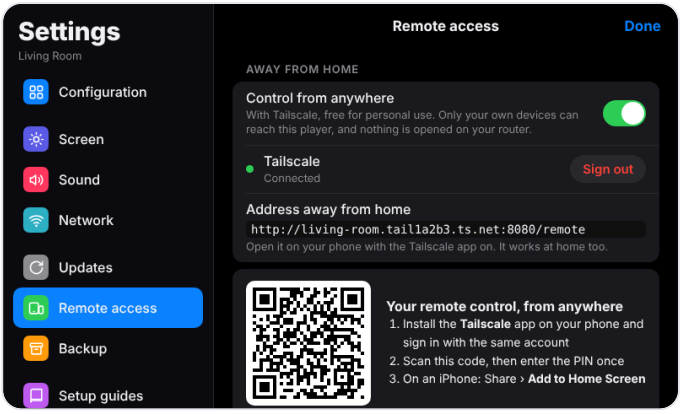
Lost the admin phone? The player's own screen can always approve devices and change levels. Tap **New PIN** there to sign every device out.

Away from home (Tailscale)

Use your remote control and Settings from anywhere: at work, on mobile data, on holiday. Tailscale connects your own devices to each other privately. It's free for personal use.

1. Settings › **Remote access** › turn on **Control from anywhere** and tap **Save**. The player installs Tailscale (a minute or two).
2. A sign-in code appears. Scan it with your phone and sign in to Tailscale (or create an account with Google, Apple, Microsoft or GitHub), then tap **Connect**. The line turns green: **Connected**.
3. On your phone, install the **Tailscale** app from the App Store or Google Play and sign in with the same account.
4. Back on the player, scan the second code (**Your remote control, from anywhere**), or open the **Address away from home** it shows. Enter the PIN once.
5. On an iPhone: Share › **Add to Home Screen**. That icon works at home and away, as long as the Tailscale app is on.





What it doesn't do: nothing is opened on your router and the player isn't on the public internet. Only devices signed in to your Tailscale account can reach it, and they still need the PIN or your admin's approval, exactly like at home.

Good to know

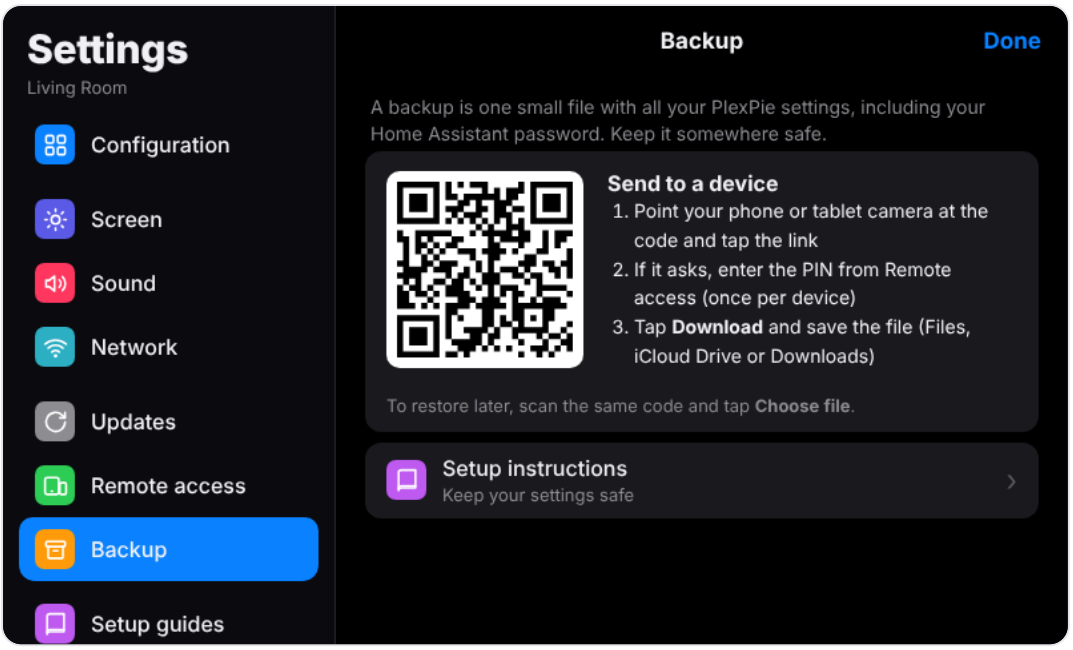
- **Several players:** turn it on for each player you want to reach away from home. The remote control's **Settings ›** button only opens players that have it.
- **Already use Tailscale?** If it was set up on this machine before PlexPie, PlexPie shows its address and leaves its settings alone. If your Tailscale reaches your whole home network (a subnet router), you don't need it on the player at all: just open the player's normal address.
- **Turn it off** to disconnect (it stays signed in), or tap **Sign out** to remove the player from your Tailscale account.
- Your Tailscale sign-in is kept by Tailscale itself, not by PlexPie, and isn't in your PlexPie backup.

Backup and restore

One small file with all your settings, for a new memory card or a second player.

Make a backup

1. On the player: Settings › **Backup**. Scan the code with your phone or tablet camera (or open Settings on a registered device).
2. Tap **Download**. On an iPhone or iPad, choose **Save to Files** (iCloud Drive is a good place).



Restore

1. Install PlexPie on the new card (Build and install).
2. Open Settings › Backup from a registered device, tap **Choose file** and pick the backup.
3. The player reinstalls itself with your settings (about a minute). Sign in to Apple Music, Spotify and Home Assistant on the screen again.

> The backup includes your Home Assistant password. Keep it somewhere private.

Troubleshooting

Most problems go away with one of the quick fixes. The full list, with commands, is in TROUBLESHOOTING on GitHub.

Quick fixes, in this order

1. **Wait two minutes** after the player starts or updates: the apps take a moment.
2. Settings › System › **Restart PlexPie** (a few seconds).
3. Settings › System › **Reboot** (about a minute).
4. Unplug the power for 10 seconds if the screen doesn't respond.
5. Settings › **Updates**: many problems are fixed in newer versions.

Problem	Try this
Home screen too big, cut off on the right	It got pinch-zoomed. Update to 2.9.2 or newer and reboot; until then, Restart PlexPie.
Plexamp screen blank or "loading"	Update to 2.9.3 or newer (it reloads itself), or Restart PlexPie.
Album art spins, music slow to start	Your router blocks `plex.direct` (DNS rebind protection). Allow `plex.direct` in its DNS settings; on pfSense add `private-domain: "plex.direct"` under Services › DNS Resolver › Custom options.
Music plays, but the Plexamp screen spins or shows no now playing	The player's address changed. Settings › Configuration › Plexamp › **Reset the Plexamp screen**, then sign in to Plex once on the screen.
Update stuck, or the version never changes	Tap Update again. On 2.8.x, update once by hand from a computer (see TROUBLESHOOTING on GitHub).
Screen jumps to Bluetooth by itself	A paired phone reconnected. Update to 2.9.2 or newer, or Forget the device.
Phone can't find the player over Bluetooth	Tap Pair a device first: it's visible for 2 minutes.
AirPlay or Spotify doesn't list the player	Same network (not guest Wi-Fi); turn on mDNS/Bonjour forwarding in the router.
Home Assistant shows "unavailable"	Wait two minutes; check the green dot in Configuration › Home Assistant.
Receiver or smart plug stops responding	Give it a fixed address in the router; unplug the receiver for a minute.
Songs skip or stutter	Remove old players with the same name on plex.tv (Account › Authorized Devices) and pick the player again in Plexamp. Then check power (official supply) and Wi-Fi; the report's playback health section shows both.
Tiles in the wrong order	Settings › Screen › **Home screen order**: move them up or down, then Save.
Screen in portrait (or upside down), but you want landscape	Settings › Screen › **Rotation**: Right or Left. Automatic follows the Raspberry Pi desktop's setting. Touch turns with it.
Screen black	Tap it (asleep). Still black: unplug for 10 seconds.
Caldera isn't in Plexamp's speaker list	Settings › Configuration › Caldera Music: check it's on and signed in. Not signed in: run the installer again from a computer (`ssh -t`). Caldera needs a 64-bit Raspberry Pi OS on a Pi 4 or 5.

Caldera plays, but other apps go silent (or the other way round)	Caldera has the DAC to itself. Set `CALDERA_DEVICE="auto"` in plexpie.conf, run the installer again and reboot.

> **Report for help:** from a computer, run `curl -fsSL https://raw.githubusercontent.com/ROGUEONE-SZ/PlexPie/main/tools/diag.sh | ssh USER@ADDRESS bash` and attach the output to an issue on GitHub.

Screen sharing

Mirror the player's screen on your Mac, iPad or iPhone and use it as if you were standing in front of it.

1. On the player: Settings › **System** › switch on **Screen sharing** and tap **Save**. It shows the address, like `vnc://192.168.1.42`.
2. **Mac:** in Finder choose **Go › Connect to Server**, type the address and click **Connect**. Enter your **Remote access PIN** as the password. Clicks work as taps.
3. **iPad or iPhone:** install the free **RealVNC Viewer** app, tap **+**, enter the address without `vnc://`, and use your PIN as the password. Tap and swipe on the picture.

Getting around without swiping

- **Home tab:** while screen sharing (or with a mouse plugged in), a small tab with a house appears on the left edge. Click it to go back to the home screen.
- **Keyboard:** press the **Home** key, or **Esc** twice.
- **Mouse:** the mouse's **Back** button also goes home.

> Screen sharing only works on your home network, and the password is your PIN: **New PIN** under Remote access changes it too. Switch it off when you don't need it.

Desktop and command line

Use the Pi for something else for a while. Music keeps playing.

1. Settings › **System** › **Exit**. The screen switches to the Raspberry Pi desktop (or the command line on Raspberry Pi OS Lite, which needs a keyboard).
2. To come back, use any of these: the **Back to PlexPie** icon on the desktop, type `plexpie` in a terminal, tap **Back to PlexPie** in Settings on your phone, or restart the player.

> With a keyboard, **Ctrl** + **Alt** + **F2** opens a text login without leaving PlexPie; **Ctrl** + **Alt** + **F1** goes back.